TOMATO ON THE VINE, large

overall

Type of crop	Vine crops
Reference	De Ruiter Seeds
Variety	Merlice
Weight	150-160 gr. (5 tomatoes per truss)

growing strategy

Growing cycle	Seasonal/year round
Production cycle	9/10 months
First production after	7 (summer)-9 (winter) weeks
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

Points of attention
BER & Bi-coloring
Very strong grower

ideal climate conditions

		DAY	NIGHT
Temperatures	Min	23°C	16°C
	Max	27°C	20°C
Relative humidity	Min	65%	80%
	Max	85%	90%
Daylength	Min	8	hours
	Max	18	hours
Radiation levels	Min	1.250	J/cm2/p.day
	Max	2.500	J/cm2/p.day

Points of attention
24 hour average temp.:
Min. 16°C without lighting
Max. 20,5°C without lighting
Min. 16,5°C with lighting
Max. 23°C with lighting

*full grown crop

technique

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

production

	WITH LIGHTING			WITHOUT LIGHTING		
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC
Min. kg/m2	-	70	70	50	60	60
Max. kg/m2	-	100	100	65	80	80

Difference between Venlo and UC occur in more extreme climates

labour

	1 ha	5 ha	10 ha	
Rest of the world	20.000	90.000	160.000	*hours*
Netherlands	10.000	45.000	80.000	*hours*

harvesting /packing

Harvesting method	manual picking/truss harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (6kg)/flow pack/plastic box
Storage	14-15°C

Boxes 5-10 kg, scales 500-750 gr.

No big temp. changes allowed, min. 12°C

other

Sensitive for viruses	No
Sensitive for Fungi	No

extra

Plant density	3 / 3,8 m2
CO2 dosing	500 / 1000 ppm

KUBO
FULL SERVICE GROW CONCEPTS

CHERRY TOMATO

overall

Type of crop	Cherry Truss
Reference	De Ruiter Seeds
Variety	Juanito
Weight	13 gr. (14 tomatoes per cluster)

growing strategy

Growing cycle	Seasonal/year round
Production cycle	9/10 months
First production after	7 (summer)-9 (winter) weeks
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

ideal climate conditions

		DAY	NIGHT
Temperatures	Min	23°C	16°C
	Max	28°C	21°C
Relative humidity	Min	60%	75%
	Max	80%	85%
Daylength	Min	8	hours
	Max	18	hours
Radiation levels	Min	1.250	J/cm2/p.day
	Max	2.500	J/cm2/p.day

technique

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

production

	WITH LIGHTING			WITHOUT LIGHTING		
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC
Min. kg/m2	-	25	25	18	23	23
Max. kg/m2	-	30	30	23	27	27

Difference between Venlo and UC occur in more extreme climates

labour

	1 ha	5 ha	10 ha	
Rest of the world	26.000	117.000	208.000	*hours*
Netherlands	13.000	58.500	104.000	*hours*

harvesting /packing

Harvesting method	manual picking/truss harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (3kg)/flow pack
Storage	14-15°C

other

Sensitive for viruses	No
Sensitive for Fungi	Yes

extra

Plant density	3,5 / 4,8 m2
CO2 dosing	500 / 1000 ppm

KUBO
FULL SERVICE GROW CONCEPTS

BEEF TOMATO

overall

Type of crop	Beef
Reference	De Ruiter Seeds
Variety	Torero
Weight	250 gr. (3/4 tomatoes per truss)

growing strategy

Growing cycle	Seasonal/year round
Production cycle	9/10 months
First production after	7 (summer)-9 (winter) weeks
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

ideal climate conditions

		DAY	NIGHT
Temperatures	Min	22,5°C	15,5°C
	Max	26,5°C	19,5°C
Relative humidity	Min	65%	80%
	Max	85%	90%
Daylength	Min	8	hours
	Max	18	hours
Radiation levels	Min	1.250	J/cm2/p.day
	Max	2.500	J/cm2/p.day

technique

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

production

	WITH LIGHTING			WITHOUT LIGHTING		
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC
Min. kg/m2	-	65	65	45	55	55
Max. kg/m2	-	85	85	60	70	70

Difference between Venlo and UC occur in more extreme climates

labour

	1 ha	5 ha	10 ha	
Rest of the world	22.000	100.000	180.000	*hours*
Netherlands	11.000	50.000	90.000	*hours*

harvesting /packing

Harvesting method	manual picking/truss harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (3kg)/flow pack
Storage	14-15°C

other

Sensitive for viruses	No
Sensitive for Fungi	Yes

extra

Plant density	2,8 / 3,5 m2
CO2 dosing	500 / 1000 ppm

KUBO
FULL SERVICE GROW CONCEPTS

CUCUMBER - LONG ENGLISH, High Wire

overall

Type of crop	High Wire
Reference	De Ruiter Seeds
Variety	Garpo
Weight	370-380 gr.

growing strategy

Growing cycle	6 months
Production cycle	5 months
First production after	1 month
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

ideal climate conditions

		DAY	NIGHT
Temperatures	Min	21°C	17,5°C
	Max	27°C	19°C
Relative humidity	Min	70%	75%
	Max	86%	82%
Daylength	Min	8	hours
	Max	18	hours
Radiation levels	Min	1.250	J/cm2/p.day
	Max	3.000	J/cm2/p.day

technique

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

production

	WITH LIGHTING			WITHOUT LIGHTING		
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC
Min. pcs/m2	220	250	280	180	220	220
Max. pcs/m2	250	320	320	230	270	270

labour

	1 ha	5 ha	10 ha	
Rest of the world	23.400	105.300	189.540	*hours*
Netherlands	11.700	52.650	94.770	*hours*

harvesting /packing

Harvesting method	manual picking/loose harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (5kg)/flow pack/plastic box/plastic seal
Storage	14-15°C

other

Sensitive for viruses	Yes
Sensitive for Fungi	Yes

KUBO
FULL SERVICE GROW CONCEPTS

CUCUMBER - LONG ENGLISH, Traditional

overall

Type of crop	Traditional (Umbrella)
Reference	De Ruiter Seeds
Variety	Gideon
Weight	400 gr.

growing strategy

Growing cycle	4 months
Production cycle	3 months
First production after	1 month
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

ideal climate conditions

		DAY	NIGHT
Temperatures	Min	21°C	17,5°C
	Max	27°C	19°C
Relative humidity	Min	70%	75%
	Max	86%	82%
Daylength	Min	8	hours
	Max	18	hours
Radiation levels	Min	1.250	J/cm2/p.day
	Max	3.000	J/cm2/p.day

technique

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

production

	WITH LIGHTING			WITHOUT LIGHTING		
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC
Min. pcs/m2	-	-	-	140	170	170
Max. pcs/m2	-	-	-	180	230	230

labour

	1 ha	5 ha	10 ha	
Rest of the world	16.600	74.700	134.460	*hours*
Netherlands	8.300	37.350	37.230	*hours*

harvesting /packing

Harvesting method	manual picking/loose harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (5kg)/flow pack/plastic box/plastic seal
Storage	14-15°C

other

Sensitive for viruses	Yes
Sensitive for Fungi	Yes

KUBO
FULL SERVICE GROW CONCEPTS

CUCUMBER - SNACK / MINI, High Wire

overall

Type of crop	High Wire
Reference	De Ruiter Seeds
Variety	Mini star
Weight	40 - 50 gr.

growing strategy

Growing cycle	6 months
Production cycle	5 months
First production after	1 month
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

ideal climate conditions

		DAY	NIGHT
Temperatures	Min	22°C	18,5°C
	Max	28°C	20°C
Relative humidity	Min	70%	75%
	Max	86%	82%
Daylength	Min	8	hours
	Max	18	hours
Radiation levels	Min	1.250	J/cm2/p.day
	Max	3.000	J/cm2/p.day

technique

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

production

	WITH LIGHTING			WITHOUT LIGHTING		
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC
Min. pcs/m2	650	800	800	485	600	600
Max. pcs/m2	800	1000	1000	600	750	750

labour

	1 ha	5 ha	10 ha	
Rest of the world	30.000	135.000	243.000	*hours*
Netherlands	15.000	67.500	121.500	*hours*

harvesting /packing

Harvesting method	manual picking/loose harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (5kg)/flow pack/plastic box
Storage	14-15°C

other

Sensitive for viruses	Yes
Sensitive for Fungi	Yes

KUBO
FULL SERVICE GROW CONCEPTS

CUCUMBER - SNACK / MINI, Traditional

overall

Type of crop	Traditional (Umbrella)
Reference	De Ruiter Seeds
Variety	Mini star
Weight	40 - 50 gr.

growing strategy

Growing cycle	4 months
Production cycle	3 months
First production after	1 month
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

ideal climate conditions

		DAY	NIGHT
Temperatures	Min	22°C	18,5°C
	Max	28°C	20°C
Relative humidity	Min	70%	75%
	Max	86%	82%
Daylength	Min	8	hours
	Max	18	hours
Radiation levels	Min	1.250	J/cm2/p.day
	Max	3.000	J/cm2/p.day

technique

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

production

	WITH LIGHTING			WITHOUT LIGHTING		
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC
Min. pcs/m2	520	640	640	388	480	480
Max. pcs/m2	640	800	800	480	600	600

labour

	1 ha	5 ha	10 ha	
Rest of the world	30.000	135.000	243.000	*hours*
Netherlands	15.000	67.500	121.500	*hours*

harvesting /packing

Harvesting method	manual picking/loose harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (5kg)/flow pack/plastic box
Storage	14-15°C

other

Sensitive for viruses	Yes
Sensitive for Fungi	Yes

KUBO
FULL SERVICE GROW CONCEPTS

ORCHIDS, Phalaenopsis

overall

Type of crop	Phalaenopsis
Variety	12 cm pots, 2 branches
Size	2/3 branches, 30 flowers, 60/80 cm

growing strategy

Growing cycle	year round
Production cycle	-
First production after	-
Origin of youngplants	yearling
Start cycle	Youngplants propagator

ideal climate conditions

		Phase 1	Phase 2	Cooling	Phase 3
Duration	Weeks	16/18	8/10	6/8	10/14
Temperatures	°C	28°C	28°C	19°C	21°C
Relative humidity	Min	65%	65%	65%	65%
	Max	75%	75%	75%	75
Daylength	Min	14	14	14	14
	Max	16	16	16	16
DLI	mol	3/3,5	4/4,5	6/7	5,5/6,5
CO_2	ppm	800	800	800	800

above 32°C is damaging

technique

Cultivation system	Open tables
Irrigation	rain system / flooded tables
Lighting	HPS / LED
Heating	Under-/above netting/gable heating

production

WITH LIGHTING

	PLASTIC	VENLO	UC
Min. pcs/m2		60	
Max. pcs/m2		65	

labour

	1 ha	5 ha	10 ha
Rest of the world			*hours*
Netherlands			*hours*

harvesting /packing

Harvesting method	table/band
Internal transport	table/system
Cleaning	no
Packaging	platic cover/trays
Type of Package	plastic/carton (/pots)
Storage	not in greenhouse / optional buffer in processing area

other

Sensitive for viruses	No	
Sensitive for Fungi	Yes	Fusarium, Rhizoctonia

KUBO
FULL SERVICE GROW CONCEPTS

ROSES

overall

Type of crop	Roses
Reference	Dümmen Orange
Variety	Avelanche
Height	30/100 cm

growing strategy

Growing cycle	year round (plant life minimum 5 years)
Production cycle	-
First production after	8 weeks
Origin of youngplants	Seeds
Start cycle	Youngplants propagator

ideal climate conditions

Temperatures	Min	16°C	
	Max	27°C	
Relative humidity	Min	65%	
	Max	80%	
Daylength	Min	18	hours
	Max	24	hours
DLI	Mol	28/30	
CO_2	ppm	800	

technique

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	HPS / LED
Heating	Heatblock/Tube rail/upper pipe

production

WITH LIGHTING

	PLASTIC	VENLO	UC
Min. pcs/m2	-	300	350
Max. pcs/m2	-	450	500

labour

	1 ha	**5 ha**	**10 ha**	
Rest of the world	30.000	120.000	240.000	*hours*
Netherlands	25.000	100.000	200.000	*hours*

7 days a week

high quality production

harvesting /packing

Harvesting method	manual cutting/ loose harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	plastic foil in buckets
Storage	cold storage

after harvesting 1 night (6°C), after packing max 1 night (2°C)

other

Sensitive for viruses	No
Sensitive for Fungi	Yes
Sensitive for Insects	Yes

Mildew

Aphid, Spider mite, Trips, Caterpillars

KUBO
FULL SERVICE GROW CONCEPTS

STRAWBERRY

overall

Type of crop	Lange dag / doordrager / Everbearer
Reference	Hoogstraten
Variety	Murano (CIV), Favori (Flevoberry)
Weight	18 / 22 gr.

growing strategy

Growing cycle	Seasonal/year round
Production cycle	8 months
First production after	12 weeks
Origin of youngplants	Runner tips
Start cycle	Tray plants

Points of attention

Harvest peaks

ideal climate conditions

		DAY	NIGHT	Points of attention
Temperatures	Min	15°C	6°C	24 hours: start, 13°C - end,17°C
	Max	25°C	12°C	
Relative humidity	Min	55%	90%	
	Max	80%	100%	
Daylength	Min	10	hours	
	Max	18	hours	
Radiation levels	Min	500	J/cm2/p.day	10 mol
	Max	1.500	J/cm2/p.day	23/24 mol

technique

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Cooling	High Pressure Fogging
Heating	Heatblock/Tube rail/grow pipe under gutter

production

	WITH LIGHTING			WITHOUT LIGHTING		
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC
Min. kg/m2	-	?	?	9	12	12
Max. kg/m2	-	?	?	13	15	15

labour

	1 ha	5 ha	10 ha	
Rest of the world				*hours*
Netherlands				*hours*

harvesting /packing

Harvesting method	Manual picking / loose harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	Manual
Type of Package	8x carton box (500 g)
Storage	3-4°C

other

Sensitive for viruses

Sensitive for Fungi

extra

Plant density	5,3 m2
CO2 dosing	800 ppm

STRAWBERRY

overall

Type of crop	Korte dag
Reference	Hoogstraten
Variety	Sonata (winter) / Sonsation (voorjaar) / Elsanta (zomer / najaar)
Weight	18 / 22 gr.

growing strategy

Growing cycle	Seasonal
Production cycle	8 / 20 weeks
First production after	4 / 12 weeks
Origin of youngplants	Runner tips
Start cycle	Tray plants

Points of attention

ideal climate conditions

		DAY	NIGHT	Points of attention
Temperatures	Min	15°C	6°C	24 hours: start, 13°C - end,17°C
	Max	25°C	12°C	
Relative humidity	Min	55%	90%	
	Max	80%	100%	
Daylength	Min	10	hours	
	Max	16	hours	
Radiation levels	Min	500	J/cm2/p.day	10 mol
	Max	1.500	J/cm2/p.day	23/24 mol

technique

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Cooling	High Pressure Fogging
Heating	Heatblock/Tube rail/grow pipe under gutter

production

	WITH LIGHTING			WITHOUT LIGHTING			
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC	
Min. kg/m2	-	?	?	4,5	3,5	5	per grow cycle, plastic has no summer cycle, no summer cycle due to time
Max. kg/m2	-	?	?	5,5	7	7	

labour

	1 ha	5 ha	10 ha	
Rest of the world				*hours*
Netherlands				*hours*

harvesting /packing

Harvesting method	Manual picking / loose harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	Manual
Type of Package	8x carton box (500 g)
Storage	3-4°C

other

Sensitive for viruses
Sensitive for Fungi

extra

Plant density	10,5 m2
CO2 dosing	800 ppm

LETTUCE, Baby Leaf

overall

Type of crop	Leafy Green
Reference	BASF
Variety	Greenflash
Weight	2-5 gr.

growing strategy

		Points of attention
Growing cycle	Year round	
Production cycle	17-19 rounds per year	min. 18 hours daylength
First production after	19-21 days	
Origin of youngplants	Seeds	
Start cycle	Direct seeding	

ideal climate conditions

		DAY	NIGHT	Points of attention
Temperatures	Min	18°C	12°C	
	Max	30°C	22°C	
Relative humidity	Min	60%	70%	
	Max	80%	90%	
Daylength	Min	12	hours	
	Max	20	hours	
DLI	Min	12	mol m2	
	Max	28	mol m2	max 700 W/m2

technique

Cultivation system	Mobile Gutter System / Deep Water Culture
Irrigation	NFT / DWC
Lighting	Yes/No
Cooling	Padwall / Mechanical cooling
Heating	Heatblock/rail pipe

production

	WITH LIGHTING			WITHOUT LIGHTING			
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC	
Min. kg/m2	-	30	35	-	15	20	
Max. kg/m2	-	50	55	-	25	30	Deep Water Culture

labour

	2 ha	4 ha	8 ha	
Rest of the world	16.640	33.280	66.560	*hours*
Netherlands	16.640	33.280	66.560	*hours*

harvesting /packing

Harvesting method	Automatic cutting
Internal transport	Conveyor system
Cleaning	No
Packaging	semi-automatic
Type of Package	flow pack/Clam shell
Storage	2-4°C

other

Sensitive for viruses	Yes	
Sensitive for Fungi	Yes	(extra: tipburn)

extra

Plant density	1200 m2 / 800 m2
CO2 dosing	500-1000 ppm (800 ppm optimal)

LETTUCE, Teen Leaf (direct seeding)

overall

Type of crop	Leafy Green
Reference	BASF
Variety	Finstar
Weight	12-20 gr.

growing strategy

		Points of attention
Growing cycle	Year round	
Production cycle	13-16 rounds per year	
First production after	22-27 days	
Origin of youngplants	Seeds	
Start cycle	Direct seeding	

ideal climate conditions

		DAY	NIGHT	Points of attention
Temperatures	Min	18°C	12°C	
	Max	28°C	18°C	
Relative humidity	Min	60%	70%	
	Max	80%	90%	
Daylength	Min	12	hours	
	Max	20	hours	
DLI	Min	12	mol m2	
	Max	28	mol m2	max 700 W/m2

technique

Cultivation system	Mobile Gutter System / Deep Float Technique
Irrigation	NFT / DFT
Lighting	Yes/No
Cooling	Padwall / Mechanical cooling
Heating	Heatblock/rail pipe

production

	WITH LIGHTING			WITHOUT LIGHTING			
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC	
Min. kg/m2	-	50	60	-	30	35	
Max. kg/m2	-	85	95	-	50	60	Mobile Gutter System

labour

	2 ha	4 ha	8 ha	
Rest of the world	16.640	33.280	66.560	*hours*
Netherlands	16.640	33.280	66.560	*hours*

harvesting /packing

Harvesting method	Automatic cutting
Internal transport	Conveyor system
Cleaning	No
Packaging	semi-automatic
Type of Package	flow pack/Clam shell
Storage	2-4°C

other

Sensitive for viruses	Yes	
Sensitive for Fungi	Yes	(extra: tipburn)

extra

Plant density	600 m2 / 250 m2
CO2 dosing	500-1000 ppm (800 ppm optimal)

LETTUCE, Teen Leaf (transplanted)

overall

Type of crop	Leafy Green	
Reference	BASF	
Variety	Multileaf	
Weight	80-120 gr.	net production 70%/80%

growing strategy

		Points of attention
Growing cycle	Year round	
Production cycle	9-10 rounds per year	
First production after	35-40 days	
Origin of youngplants	Seeds	
Start cycle	Germination / propagation	

ideal climate conditions

		DAY	NIGHT	Points of attention
Temperatures	Min	18°C	12°C	
	Max	28°C	18°C	
Relative humidity	Min	60%	70%	
	Max	80%	90%	
Daylength	Min	9	hours	
	Max	20	hours	
DLI	Min	12	mol m2	
	Max	28	mol m2	max 700 W/m2

technique

Cultivation system	Mobile Gutter System / Deep Float Technique	
Irrigation	NFT / DFT	
Lighting	Yes/No	
Cooling	Padwall / Mechanical cooling	
Heating	Heatblock/rail pipe	

production

	WITH LIGHTING			WITHOUT LIGHTING			
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC	
Min. kg/m2	-	50	60	-	30	35	
Max. kg/m2	-	85	95	-	50	60	Mobile Gutter System

labour

	2 ha	4 ha	8 ha	
Rest of the world	16.640	33.280	66.560	*hours*
Netherlands	16.640	33.280	66.560	*hours*

harvesting /packing

Harvesting method	Automatic cutting
Internal transport	Conveyor system
Cleaning	No
Packaging	semi-automatic
Type of Package	flow pack/Clam shell
Storage	2-4°C

other

Sensitive for viruses	Yes	
Sensitive for Fungi	Yes	(extra: tipburn)

extra

Plant density	500 m2 / (30-45) m2
CO2 dosing	500-1000 ppm (800 ppm optimal)

LETTUCE, Whole head (medium)

overall

Type of crop	Butter head	
Reference	BASF	
Variety	Matthew	
Weight	150-250 gr.	

growing strategy

		Points of attention
Growing cycle	Year round	
Production cycle	8-9 rounds per year	
First production after	40-45 days	
Origin of youngplants	Seeds	
Start cycle	Germination / propagation	Transplanting

ideal climate conditions

		DAY	NIGHT	Points of attention
Temperatures	Min	18°C	12°C	Optimal lower than Baby leaf
	Max	28°C	18°C	
Relative humidity	Min	50%	60%	
	Max	80%	90%	
Daylength	Min	12	hours	
	Max	18	hours	
DLI	Min	12	mol m2	
	Max	28	mol m2	max 700 W/m2

technique

Cultivation system	Mobile Gutter System / Deep Float Technique	
Irrigation	NFT / DFT	
Lighting	Yes/No	
Cooling	Padwall / Mechanical cooling	
Heating	Heatblock/rail pipe	

production

	WITH LIGHTING			WITHOUT LIGHTING			
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC	
Min. kg/m2	-	30	35	-	20	25	
Max. kg/m2	-	55	60	-	40	45	Mobile Gutter System

labour

	2 ha	4 ha	8 ha	
Rest of the world	41.600	83.200	166.400	*hours*
Netherlands	41.600	83.200	166.400	*hours*

harvesting /packing

Harvesting method	Manual / semi-automatic harvesting
Internal transport	Conveyor system / trolleys
Cleaning	Fresh water spraying (before harvesting)
Packaging	Manual / semi-automatic
Type of Package	Sleeves / boxes (bulk)
Storage	2-4°C

other

Sensitive for viruses	Yes	
Sensitive for Fungi	Yes	(extra: tipburn)

extra

Plant density	350 m2 / (20-25) m2
CO2 dosing	500-1000 ppm (800 ppm optimal)

LETTUCE, Whole head (large)

overall

Type of crop	Butter head	
Reference	BASF	
Variety	Clarke	
Weight	250-350 gr.	

growing strategy

		Points of attention
Growing cycle	Year round	
Production cycle	6-8 rounds per year	
First production after	45-58 days	
Origin of youngplants	Seeds	
Start cycle	Germination / propagation	Transplanting

ideal climate conditions

		DAY	NIGHT	Points of attention
Temperatures	Min	18°C	12°C	Optimal lower than Baby leaf
	Max	28°C	18°C	
Relative humidity	Min	50%	60%	
	Max	80%	90%	
Daylength	Min	12	hours	
	Max	18	hours	
DLI	Min	12	mol m2	
	Max	28	mol m2	max 700 W/m2

technique

Cultivation system	Mobile Gutter System / Deep Float Technique	
Irrigation	NFT / DFT	
Lighting	Yes/No	
Cooling	Padwall / Mechanical cooling	
Heating	Heatblock/rail pipe	

production

	WITH LIGHTING			WITHOUT LIGHTING			
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC	
Min. kg/m2	-	30	35	-	20	25	
Max. kg/m2	-	55	60	-	40	45	Mobile Gutter System

labour

	2 ha	4 ha	8 ha	
Rest of the world	41.600	83.200	166.400	*hours*
Netherlands	41.600	83.200	166.400	*hours*

harvesting /packing

Harvesting method	Manual / semi-automatic harvesting
Internal transport	Conveyor system / trolleys
Cleaning	Fresh water spraying (before harvesting)
Packaging	Manual / semi-automatic
Type of Package	Sleeves / boxes (bulk)
Storage	2-4°C

other

Sensitive for viruses	Yes	
Sensitive for Fungi	Yes	(extra: tipburn)

extra

Plant density	250 m2 / (15-20) m2
CO2 dosing	500-1000 ppm (800 ppm optimal)

CANNABIS

overall

Type of crop	Short day crop
Variety	
Size / weight	80 grams per 1 meter plant

growing strategy

Growing cycle	year round
Production cycle	
First production after	14 weeks
Origin of youngplants	Seeds
Start cycle	Clones

ideal climate conditions

		Cloning	Veg Time	Flowering	Drying	Lab testing
Duration	Weeks	3	2	8	10 days	2
Temperatures	Min	20°C	18°C	18°C		
	Max	24°C	25°C	27°C		
Relative humidity	Min	70%	70%	70%	55%	
	Max	95%	85%	85%	70%	
Daylength	Min	18	18	10		
	Max	20	20	12		
DLI	mol	25	25	25	25	25
CO_2	ppm	600/800	600/800	600/800	600/800	600/800

technique

Cultivation system	Tables (containers)
Irrigation	Flood / Drain
Lighting	Yes
Cooling	Mechanical cooling (dehumidification)
Heating	Heatblock

production

	WITH LIGHTING		
	PLASTIC	VENLO	UC
Min. pcs/m2	0,7	1	1,5
Max. pcs/m2	1,2	1,5	2,2

labour

	1 ha	5 ha	10 ha	
Rest of the world	62.400	312.000	624.000	*hours*
Netherlands	-	-	-	*hours*

harvesting /packing

Harvesting method	Manual Picking
Internal transport	Table system
Cleaning	Not applicable
Packaging	Semi-automatic
Type of Package	Jars (3,5 gram), Bulk packaging (2kg)
Storage	Fault (16 degrees, low humidity <55%)

other

Sensitive for viruses	No	Molt, Powder Mildew
Sensitive for Fungi	Yes	

extra

Plant density	flowering (4,5 m2)

KUBO
FULL SERVICE GROW CONCEPTS

SWEET POINT PEPPER

overall

Type of crop	High wire crop
Reference	Rijk Zwaan
Variety	Palermo
Weight	125 gram

growing strategy

Growing cycle	1 cycle a year
Production cycle	50 weeks
First production after	15 weeks
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

Points of attention

ideal climate conditions

		DAY	NIGHT	
Temperatures	Min	21°C	16°C	
	Max	27°C	21°C	
Relative humidity	Min	75%	80%	
	Max	80%	85%	
Daylength	Min	-	hours	
	Max	16	hours	
Radiation levels	Min	1.000	J/cm2/p.day	1.500 optimal
	Max	2.000	J/cm2/p.day	

Points of attention

technique

Cultivation system	Grow gutter
Irrigation	drip
Lighting	Possible
Cooling	HPF/padwall/mechanical cooling
Heating	heat block/rail pipe

production

	WITH LIGHTING			WITHOUT LIGHTING		
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC
Min. kg/m2	?	20	20	?	18	18
Max. kg/m2	?	24	24	?	22	22

labour

	1 ha	5 ha	10 ha	
Rest of the world	20.800	104.000	208.000	*hours*
Netherlands	10.400	52.000	104.000	*hours*

harvesting /packing

Harvesting method	manual picking
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	flow pack
Storage	14-15°C

other

Sensitive for viruses	No
Sensitive for Fungi	No

extra

Plant density	9 per m2
CO2 dosing	600 - 800 ppm

BELL PEPPER

overall

Type of crop	High wire crop
Reference	Rijk Zwaan
Variety	Red Line
Weight	210 gram

growing strategy

Growing cycle	1 cycle a year
Production cycle	50 weeks
First production after	15 weeks
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

Points of attention

ideal climate conditions

		DAY	NIGHT	
Temperatures	Min	21°C	18°C	
	Max	27°C	21°C	
Relative humidity	Min	75%	80%	
	Max	80%	85%	
Daylength	Min	-	hours	
	Max	16	hours	
Radiation levels	Min	1.000	J/cm2/p.day	1500 optimal
	Max	2.000	J/cm2/p.day	

Points of attention

technique

Cultivation system	Grow gutter
Irrigation	drip
Lighting	Possible
Cooling	HPF/padwall/mechanical cooling
Heating	heat block/rail pipe

production

	WITH LIGHTING			WITHOUT LIGHTING		
	PLASTIC	VENLO	UC	PLASTIC	VENLO	UC
Min. kg/m2	?	36	36	?	32	32
Max. kg/m2	?	42	42	?	38	38

labour

	1 ha	5 ha	10 ha	
Rest of the world	20.800	104.000	208.000	*hours*
Netherlands	10.400	52.000	104.000	*hours*

harvesting /packing

Harvesting method	manual picking
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (5kg)/flow pack/plastic box
Storage	14-15°C

other

Sensitive for viruses	No
Sensitive for Fungi	No

extra

Plant density	7/ 7,5 per m2
CO2 dosing	600 - 800 ppm